

Seongjun Kim, Ph.D.

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Email: 2sjkim@gmail.com

Phone: (+82) 10-8925-7834

EDUCATION

Yonsei University

Seoul, Korea

Ph.D., School of Integrated Technology, College of Engineering

2020.03-2026.02

Advisor: Prof. Jongduk Baek

Thesis: Deep learning-based artifact reduction algorithms for low-dose CT images

Yonsei University

Seoul, Korea

B.S., School of integrated Technology, College of Engineering

2017.03-2020.02

High Honors in Graduation (Ranked 1st in class)

Sejong Science High School

Seoul, Korea

2014.03-2017.02

RESEARCH INTERESTS

My research interests lie in deep learning (DL)-based image enhancement, particularly in the context of medical imaging systems. During my Ph.D. studies, I have focused on developing artifact reduction algorithms for the ill-posed problem of computed tomography (CT). Based on these interests, I aim to contribute image enhancement technologies to diverse imaging modalities and real-world industrial applications. Specifically, I have explored the following research areas.

- **Data complexity:** The intrinsic complexity of data caused by multiple artifact types poses a fundamental challenge for DL models, often leading to unstable network training and image quality degradation. I have designed DL framework to address severe artifact patterns under highly ill-posed CT settings.
- **Self-supervised learning:** The dependence of DL models on a large-scale paired dataset is one of the main problems that restrict their applicability in real imaging systems. I have focused on self-supervised DL framework for artifact reduction that constructs training dataset without reference data, which can make this pipeline suitable for deployment in data-constrained clinical and industrial environments.
- **Generalizability:** DL models typically exhibit limited generalizability across diverse clinical scenarios, such as scanning protocols, data acquisition parameters, and noise levels. I have developed a generalizable denoising model for unseen image of arbitrary noise level.

WORK EXPERIENCE

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| Yonsei University | <i>Seoul, Korea</i> |
| • Postdoctoral Scholar | <i>2026.03-Current</i> |
| • Research Intern | <i>2019.03-2020.02</i> |
| - Worked on deep learning-based CT image quality enhancement | |
| - Advisor: Prof. Jongduk Baek | |

PUBLICATIONS

1. Convolutional neural network-based metal and streak artifacts reduction in dental CT images with sparse-view sampling scheme
Seongjun Kim*, Junhyun Ahn*, Byeongjoon Kim*, Chulhong Kim, Jongduk Baek
Medical Physics, 2022
2. Sparsier2Sparse: Self-supervised convolutional neural network-based streak artifacts reduction in sparse-view CT images
Seongjun Kim, Byeongjoon Kim, Jooho Lee, Jongduk Baek
Medical Physics, 2023
3. Dose-aware denoising diffusion model for low-dose CT
Seongjun Kim*, Byeongjoon Kim*, Jongduk Baek
Physics in Medicine & Biology, 2025
4. X-ray CT metal artifact reduction using neural attenuation field prior
Jooho Lee, **Seongjun Kim**, Junhyun Ahn, Adam S. Wang, Jongduk Baek
Medical Physics, 2025

CONFERENCE

1. A hybrid domain approach to reduce streak artifacts of sparse view CT image via convolutional neural network
Seongjun Kim, Byeongjoon Kim, Jongduk Baek
SPIE Medical Imaging, 2021
2. Convolutional neural network-based metal artifact correction for sparse view dental CT imaging
Seongjun Kim*, Junhyun Ahn*, Jongduk Baek
AAPM, 2021 (*Oral presentation*)
3. Sparsier2Sparse: Weakly-supervised learning for streak artifact reduction with unpaired sparse-view CT data
Seongjun Kim, Byeongjoon Kim, Jongduk Baek
CT Meeting, 2022 (*Oral presentation*) / *IFMIA* 2023

4. Dose-aware adaptive denoising network for low-dose CT
Seongjun Kim, Byeongjoon Kim, Jongduk Baek
SPIE Medical Imaging, 2025
5. A method to reduce truncation artifacts in dental CT with a offset detector with sinogram domain learning
 Taijun Kang, **Seongjun Kim**, Jongduk Baek
Fully 3D, 2025

AWARDS AND SCHOLARSHIP

- **Best Presentation Award** 2023
The International Forum on Medical Imaging in Asia (IFMIA)
- **Certificate of Commendation** | Incheon 2020
Awarded for outstanding academic achievement
- **Academic Honors in Engineering** | Yonsei University 2019-1, 2019-2
- **Full scholarship for Undergraduate School** | Yonsei university 2017.03-2020.02
Institute for Information and Communications Technology Promotion (IITP)

TEACHING EXPERIENCE

- Teaching Assistant** (Yonsei University)
- AAI7101 Medical Imaging System and Its Application for Artificial Intelligence 2022-2
 - IIT7011 Introduction to Bioimaging 2020-2
 - IIT3012 Discrete Signal Processing 2020-2
 - IIT6022 The Fourier Transform and Its Applications 2020-1
 - IIT2102 Advanced Mathematics II 2020-1

SKILLS

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| Programming Languages | Python, Matlab |
| Machine Learning Tools | PyTorch, Tensorflow, OpenCV, NumPy, Scikit-learn |

Reviewer

Physics in Medicine & Biology (Achieved IOP Trusted Reviewer status), MICCAI, Engineering Applications of Artificial Intelligence, Biomedical Signal Processing and Control

